Pairing-Based Registered Attribute-Based Encryption

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by

Golla Meghanandh Manvith Prabhash (22CS30027)

Under the supervision of **Dr. Somindu Chaya Ramanna**

Department of Computer Science and Engineering
Indian Institute of Technology Kharagpur

Spring Semester, 2025-26
April 2026

DECLARATION

I certify that

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
KHARAGPUR - 721302, INDIA

CERTIFICATE

This is to certify that the project report entitled **"Pairing-Based Registered Attribute-Based Encryption"** submitted by **Golla Meghanandh Manvith Prabhash (Roll No: 22CS30027)** to the Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of the degree of **Dual Degree (B. Tech + M. Tech)** in **Computer Science and Engineering** is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2025-26.

Date: April 10, 2026
Place: Kharagpur

Dr. Somindu Chaya Ramanna
Department of Computer Science and Engineering
Indian Institute of Technology Kharagpur
Kharagpur - 721302, India

Abstract

Name of the student: Golla Meghanandh Manvith Prabhash **Roll No:** 22CS30027
Degree for which submitted: Dual Degree (B. Tech + M. Tech)
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Attribute-Based Encryption (ABE) extends conventional public-key systems, permitting data access to be governed by specific policies. The drawback of classic ABE is its dependence on a single, completely trusted central authority for key issuance. Should an attacker breach this central node, the resulting key escrow vulnerability means every encrypted message in the entire network becomes exposed. In the previous semester, we investigated Identity-Based Encryption (IBE) over the Computational Diffie-Hellman (CDH) assumption to solve this. While theoretically sound, the sheer computational weight of its underlying garbled circuits made it unusable in practice. Therefore, the focus of this report shifts to a highly scalable, practical solution: Registered Attribute-Based Encryption (Reg-ABE). By dispensing with a master authority, Reg-ABE introduces a non-privileged, deterministic key curator. Users locally derive their own secret keys. They then forward only their public keys and verified attributes to the curator for registration. This thesis details the aggregation processes involved in pairing-based Reg-ABE setups. It explores how converting composite-order groups into prime-order domains significantly improves efficiency. Additionally, the report looks into recent functional extensions, specifically Registered Functional Encryption (Reg-FE), which enables inner-product and quadratic computations directly on encrypted payloads. Finally, the text explores fundamental cryptographic principles, highlights open questions in the functional encryption space, and sets a trajectory for future software implementations and post-quantum preparedness.

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Abbreviations

ABE	Attribute-Based Encryption
CCA	Chosen-Ciphertext Attack
CP-ABE	Ciphertext-Policy Attribute-Based Encryption
IBE	Identity-Based Encryption
RBE	Registration-Based Encryption
Reg-ABE	Registered Attribute-Based Encryption
Reg-FE	Registered Functional Encryption
Reg-IPFE	Registered Inner-Product Functional Encryption
Reg-QFE	Registered Quadratic Functional Encryption
CDH	Computational Diffie-Hellman (Assumption)
DDH	Decisional Diffie-Hellman (Assumption)
k-Lin	k-Linear (Assumption)
LSSS	Linear Secret Sharing Scheme
PKC	Public-Key Cryptography
PKG	Private Key Generator
GC	Garbled Circuit
CRS	Common Reference String
DPVS	Dual Pairing Vector Spaces

Chapter 1

Introduction

The widespread adoption of decentralized storage and third-party cloud computing has forced a re-evaluation of how sensitive information is secured. Legacy encryption protocols usually focus on direct, point-to-point connections and struggle to support the complex, dynamic permission structures required by large organizations. Public-key cryptography (PKC) forms the basis of secure digital communication; however, as the number of users and the complexity of data relations grow, the limitations of identity-based management become apparent. Attribute-Based Encryption (ABE), introduced by Sahai and Waters in 2005 [7], generalizes PKC by embedding access policies directly into the cryptographic framework. In a Ciphertext-Policy ABE (CP-ABE) scheme, an encryptor associates a ciphertext with a boolean access policy (e.g., Director $\vee$ (Manager $\wedge$ HR)). A user can only decrypt the ciphertext if their cryptographically certified attributes satisfy the formula. This "one-to-many" encryption approach allows for fine-grained access control without requiring the encryptor to know the specific identities of the recipients.

1.1 The Key Escrow Bottleneck

Despite its utility for cloud storage, electronic medical records (EMR), and large-scale organizational data sharing, traditional ABE alters the trust model of standard PKC. It mandates a central trusted authority—the Private Key Generator (PKG)—to manage the system's master secret parameters. The core of the "key escrow" problem lies in the key generation phase. To issue a functional secret key to a user, the PKG must use a Master Secret Key (MSK). Mathematically, if SK_{attr} is the user's key and f is the key generation function:

$$SK_{attr} \leftarrow f(MSK, \text{Attributes}) \tag{1.1}$$

Consequently, the PKG holds a long-term master secret key capable of decrypting *every* ciphertext in the system. This creates a single point of failure: if an adversary compromises this central authority, or if the authority itself is malicious (the "honest-but-curious"

or "malicious provider" model), the privacy of every user is permanently revoked. This vulnerability remains the most significant barrier to the widespread real-world deployment of ABE and Identity-Based Encryption (IBE) frameworks in zero-trust environments.

1.2 The Shift to Registered ABE

To fundamentally eliminate the key escrow problem, the cryptographic community has recently shifted toward "Registration-Based" primitives. Following the introduction of Registration-Based Encryption (RBE) for identities by Garg et al. (2018) [3], Hohenberger, Lu, Waters, and Wu [4] generalized this concept to Registered Attribute-Based Encryption (Reg-ABE). In a Reg-ABE system, the absolute trusted authority is completely replaced by a transparent "key curator." The structural advantages of this shift are three-fold:

1. **Client-Side Key Generation:** Rather than a central server generating secrets, users compute their own (sk_u, pk_u) pairs locally. This process closely mirrors standard key generation seen in RSA or elliptic curve setups. As a result, the curator is completely unaware of any user's sk_u.

2. **Trustless Aggregation:** The curator does not store or manage secret parameters. Its sole function is to execute a deterministic algorithm that bundles all submitted public keys into a single Master Public Key (MPK). Alongside this, it publishes Helper Secret Keys (HSK) which users need to process their decryptions.

3. **Post-Compromise Security:** Even if the curator's entire database is leaked, no user data is compromised because the curator never possessed the "root" decryption capability.

1.3 Project Scope and Outline

During the previous semester, this project focused on simulating IBE from the standard Computational Diffie-Hellman (CDH) assumption to bypass escrow using "non-black-box" techniques. While theoretically sound, these methods involved complex circuit garbling and heavy cryptographic machinery that proved practically infeasible for realistic workloads. By switching to bilinear pairings and algebraic curves, this research intends to preserve the escrow-free nature of the system while substantially reducing ciphertext overhead and accelerating computation times. The remainder of this document follows the structure below:

- **Chapter 2:** Provides a recap of the previous semester's work, explaining the severe performance limitations of CDH-based garbled simulations that prompted the pivot to pairings.

- **Chapter 3:** Introduces the core mathematical toolkit, defining cyclic groups, bilinear pairings, and the mechanics of secret sharing.
- **Chapter 4:** Breaks down the mathematical procedures that drive pairing-based Reg-ABE, with a detailed analysis of the attribute registration mechanism.
- **Chapter 5:** Analyzes the computational leaps gained by transitioning cryptographic operations from bulky composite-order groups to agile prime-order spaces.
- **Chapter 6:** Reviews modern iterations of Registered Functional Encryption (Reg-FE), demonstrating how calculations can be run over hidden data.
- **Chapter 7:** Summarizes existing roadblocks in the field and proposes a roadmap for implementing these protocols in decentralized environments.

Chapter 2

IBE from CDH and the Wall of Infeasibility

To understand the motivation behind using bilinear pairings for Registered Attribute-Based Encryption (Reg-ABE), we must first look at the results of Project-I. That initial research phase tackled a specific problem: bypassing key escrow by constructing an Identity-Based Encryption (IBE) protocol that relied entirely on the standard Computational Diffie-Hellman (CDH) assumption.

2.1 The Non-Black-Box Simulation

Historically, constructing IBE required bilinear pairings or lattice-based cryptography. The Döttling-Garg construction [1] bypassed traditional impossibility results by embedding the logic of a novel primitive—Chameleon Encryption—inside a deeply nested chain of Garbled Circuits (GCs). During the first phase of the project (up to the BTP-1 presentation), we successfully built a functional simulation of this theoretical framework. Instead of strictly compiling complex cryptographic field mathematics into billions of boolean gates, we implemented a "decryption-as-path-verification" mechanism. By validating logic through chameleon hash collisions rather than monolithic circuit compilation, we proved that CDH assumptions alone could theoretically support an escrow-free IBE, bypassing the need for a central trapdoor.

2.2 Post-BTP-1 Optimizations: Tree Arity and AES Labeling

Following the BTP-1 presentation, our primary focus shifted to optimizing the garbled circuit chains, which were severely bottlenecking the system. In the naive Döttling-Garg construction, processing an n-bit identity (e.g., 512 bits) via a binary tree structure

requires a depth of 512 sequential garbled circuits, leading to an exponential expansion in ciphertext size. To mitigate this, we introduced a **Tree Arity Optimization**. Instead of processing the identity bit-by-bit, we restructured the evaluation tree to process the identity in larger blocks, significantly reducing the depth of the circuit chain.

2.2.1 Label Chunking and AES Integration

As derived from our optimization models, we reduced the standard 512-bit monolithic label requirement by chunking the identity space. Specifically, an identity was broken down into manageable 128-bit segments (e.g., 4×128 bits for a 512-bit ID). Let $\tilde{\Pi}_i$ represent the state at depth i of the tree traversal. Instead of relying purely on heavy garbled evaluations for state transitions, we utilized the active label $\tilde{\Pi}_i$ to derive a symmetric key K. We then used standard AES encryption to secure the traversal path to the next label:

$$K \leftarrow \mathrm{KDF}(\tilde{\Pi}_i) \tag{2.1}$$

$$C_{out} = \mathrm{AES}_K(\text{Next Label}) \tag{2.2}$$

The final ciphertext ct at each node became a hybrid of this AES output and the minimal necessary Garbled Circuit (GC) payload:

$$ct = C_{out} + GC \tag{2.3}$$

By utilizing 128-bit wire labels instead of 512-bit labels, and bridging the tree nodes with highly optimized AES blocks, we substantially reduced the memory footprint of the simulation. Our empirical findings concluded that *"128 bits is strictly better than 512 bits"* for the garbled wire parameters, as it provided standard security while keeping the GC evaluation times within a testable threshold.

2.3 The Practical Infeasibility of Garbled Cryptography

Despite these arity and symmetric-key optimizations, our ultimate analysis demonstrated that the non-black-box technique remains practically infeasible for real-world deployment. The fundamental limitation of the CDH-based IBE scheme is that at the core of the evaluation, the garbled circuit must still process mathematical operations over large prime fields. Translating operations like elliptic curve scalar multiplications into boolean circuits for garbling yields extreme inefficiencies.

Even when leveraging 128-bit segment optimizations, translating mathematical fields into boolean gates inflates the resulting garbled output to multi-megabyte capacities per transmission. This extreme demand on processing time and storage inherently disqualifies

Table 2.1: Estimated Cost of Cryptographic Operations as Boolean Circuits

Operation	**Native Cost**	**Cost as Boolean Circuit (Gates)**
Modular Exponentiation / Elliptic Curve Scalar Mult.	≈ 1 group op.	$10^6 - 10^9$ or more
Field Multiplication	≈ 1 field op.	Thousands of gates
AES-128 (Our Optimization)	Hardware Accel.	$\approx 3,000 - 5,000$ gates

the method from practical application in cloud or dynamic network scenarios.

Due to the extreme computational overhead and practical infeasibility of this approach, the remaining of this research focuses on a fundamentally different cryptographic framework.

Chapter 3

Primitives

This section maps out the fundamental algebraic elements required to understand the protocols discussed later in this report, covering cyclic groups, pairings, vector spaces, and secret sharing schemes.

3.1 Groups and Hardness Assumptions

In mathematics, a group $\mathbb{G}$ is a designated set of elements that, when combined via a specific operation (usually addition or multiplication), obey the laws of closure, associativity, identity, and invertibility. The cryptographic field relies heavily on cyclic groups restricted to a prime order p. Such groups contain a generator element g, which can be manipulated to produce any other element in the group, written as g^a for some integer $a \in \mathbb{Z}_p$. The underlying security of these structures assumes that solving the Discrete Logarithm Problem (DLP) is computationally out of reach. Specifically, if an attacker is given g and g^a, it is assumed impossible for them to efficiently deduce the exponent a. When dealing with advanced pairing architectures, these security proofs are extended into more rigorous assumptions:

- **k-Linear (k-Lin) Assumption:** This is a generalization of the Decisional Diffie-Hellman (DDH) assumption. It posits that if an observer is given a set of vectors located within an affine subspace of dimension k, they lack the computational power to determine whether a newly introduced challenge vector belongs to that exact subspace or if it was just sampled randomly from the entire vector space. This specific property is paramount for proving semantic security in prime-order Reg-ABE schemes.

3.2 Bilinear Pairings

A bilinear pairing serves as an algorithmic bridge, taking elements from two distinct cryptographic groups and mapping them into a third. Let us assume two cyclic groups, $\mathbb{G}_1$ and $\mathbb{G}_2$, both of prime order p and possessing generators g_1 and g_2. We also define a target group $\mathbb{G}_T$ of the same order. A pairing is mathematically expressed as a map $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ that meets the following three conditions:

1. **Bilinearity:** For any given elements $u \in \mathbb{G}_1$ and $v \in \mathbb{G}_2$, and any chosen scalars $a, b \in \mathbb{Z}_p$, the mapping must satisfy $e(u^a, v^b) = e(u, v)^{ab}$.

2. **Non-degeneracy:** When mapping the foundational generators together, the result $e(g_1, g_2)$ must not equal 1 (which represents the identity element of the target group $\mathbb{G}_T$). This ensures the map produces mathematically significant outputs rather than collapsing to a useless default state.

3. **Computability:** An efficient, polynomial-time algorithm must exist to calculate $e(u, v)$ for all valid inputs.

This property of bilinearity is the core mechanism of ABE; it gives users the mathematical ability to perform a single multiplicative operation while keeping the actual values hidden inside the exponents.

3.2.1 Symmetric vs. Asymmetric Pairings

Cryptographers categorize pairings into three distinct types depending on how their base groups ($\mathbb{G}_1$ and $\mathbb{G}_2$) relate to one another:

- **Type I (Symmetric):** In this setup, both base groups are identical ($\mathbb{G}_1 = \mathbb{G}_2$). While these symmetric models make it easy to design new protocols, they force the system to use substantially larger key sizes to stay secure, which severely degrades operational performance.

- **Type II (Asymmetric):** Here, a computable homomorphism $\phi : \mathbb{G}_2 \to \mathbb{G}_1$ exists in one direction, but calculating the reverse homomorphism is mathematically intractable.

- **Type III (Asymmetric):** This type possesses no efficient homomorphism connecting $\mathbb{G}_1$ and $\mathbb{G}_2$ in either direction. Modern cryptographic deployments rely almost exclusively on Type III pairings because they allow for exceptionally compact parameter sizes (usually built on Barreto-Naehrig or BLS12-381 curves). This translates directly to faster compute speeds and much smaller ciphertexts.

3.3 Composite-Order Groups and Subgroup Orthogonality

While prime-order groups are the standard for modern efficiency, many foundational cryptographic protocols were originally designed using *composite-order* bilinear groups. In a composite-order setting, the total size of the group (its order N) is a product of distinct primes (for example, $N = p_1 p_2 p_3$).

By applying the Chinese Remainder Theorem, a composite group of order N can be mathematically decomposed into independent, non-overlapping subgroups $\mathbb{G}_{p_1}, \mathbb{G}_{p_2}$, and $\mathbb{G}_{p_3}$.

The most critical feature of composite-order groups is the property of **subgroup orthogonality**. If a bilinear pairing e is applied to elements from two different subgroups, the output is mathematically forced to be the identity element (1) of the target group. For any $h_1 \in \mathbb{G}_{p_1}$ and $h_2 \in \mathbb{G}_{p_2}$:

$$e(h_1, h_2) = 1 \tag{3.1}$$

This orthogonality provides an incredibly intuitive mechanism for access control and decryption. An encryptor can place the actual cryptographic payload (the message) in one subgroup (e.g., p_1) and place random blinding factors or security noise in another subgroup (e.g., p_2). During decryption, when the pairing operation is executed by an authorized user, the orthogonal subgroups interact to produce 1. This effectively cancels out the random noise completely, leaving only the clear data behind. However, the large parameter sizes required to keep the factorization of N secure make composite-order groups highly inefficient in practice.

3.4 Linear Secret Sharing Schemes (LSSS)

In order to enforce precise, fine-grained access policies within an ABE system, raw boolean logic must be mathematically converted into matrices. This conversion is handled by a Linear Secret Sharing Scheme (LSSS). Under this scheme, an access policy takes the form of a share-generating matrix M with dimensions $l \times n$, coupled with a function ρ that maps individual rows of M to specific attributes. When an encryptor wants to hide a secret s, they generate a column vector $\vec{v} = (s, r_2, \ldots, r_n)^T$, filling r_i with random values. The shares of that secret are defined by the elements resulting from the multiplication $M \cdot \vec{v}$. During the decryption phase, if the decryptor's attributes satisfy the required access structure, they can derive constants c_i to solve $\sum c_i(M_i \cdot \vec{v}) = s$. Reconstructing this equation enables the user to strip the blinding factors away from the encrypted payload. Recent updates to Reg-ABE [6] have expanded LSSS capabilities to include **non-monotone span programs**. This means access policies can now dictate negative

requirements (for example, allowing access to someone who is a Director $\wedge$ $\neg$Contractor). To accomplish this, the cryptographic span program must be engineered to mathematically represent NOT-gates within the linear algebra of the underlying vector spaces.

3.5 Dual Pairing Vector Spaces (DPVS)

Dual Pairing Vector Spaces supply the mathematical translations needed to replicate the behavior of composite-order groups within highly efficient prime-order systems. Drawing on the foundational work by Lewko (2011) [5], DPVS leverages orthogonal vector bases to emulate the independent "canceling property" normally provided by composite subgroups.

3.5.1 The Mechanism of Orthogonality

At initialization, the setup algorithm generates two distinct bases for the vector spaces, denoted as $\mathbb{B}$ and $\mathbb{B}^*$. These bases are engineered to be strictly orthogonal under the bilinear pairing operation.

Let $b_i \in \mathbb{B}$ and $b_j^* \in \mathbb{B}^*$. Due to their orthogonal relationship, calculating the dot-product pairing of these basis vectors adheres to the following rule:

$$e(b_i, b_j^*) = \begin{cases} g_T^{\psi} & \text{if } i = j \\ 1 & \text{if } i \neq j \end{cases} \tag{3.2}$$

When $i \neq j$, the pairing maps the vectors perfectly to the target group's identity element (1), effectively erasing them from the mathematical equation.

3.5.2 Simulating Subgroups for Decryption

This strict mathematical orthogonality is utilized to isolate sensitive data from cryptographic noise. In a Reg-ABE scheme, the encryptor places the actual ciphertext payload into a specific subset of the basis vectors (e.g., b_1, b_2), while placing random blinding factors—the "noise" used to secure the data against unauthorized users—into a separate subset of the vectors (e.g., b_3, b_4).

During decryption, an authorized user applies their secret key (which resides in the dual basis $\mathbb{B}^*$). Because of the orthogonal design, the pairing algorithm naturally isolates the payload while perfectly eliminating the random noise (the blinding factors). This accurately mirrors the subgroup isolation techniques of older composite-order schemes, allowing the system to securely aggregate user traits while operating entirely within the boundaries of fast, prime-order elliptic curves.

3.6 Vector Commitments and Merkle Trees

In a purely decentralized Reg-ABE network, the Key Curator faces a significant storage and communication bottleneck: it must track the public keys of all participants without creating a massive, continuously expanding string of parameters. If the Master Public Key (MPK) grew linearly with every new user, the system would quickly become too bloated for practical cloud deployment.

3.6.1 The Compression Mechanism

Vector commitments solve this scalability issue by mathematically condensing an ordered list of registered values $(x_1, \ldots, x_N)$ into a fixed-size cryptographic footprint. These compressed values serve as the external leaves of a **Merkle Hash Tree**.

Whenever users register, their public keys are hashed. Sibling nodes in the tree are concatenated and hashed together ($H(\text{Left} \parallel \text{Right})$), a process that recursively bubbles upward through the tree's layers until it terminates at a single, unified Root Hash. This definitive, constant-size root acts as the system's global Master Public Key (MPK).

3.6.2 Logarithmic Verification (Merkle Proofs)

The primary advantage of this structure is that it allows the system to remain completely "trustless." If an individual user wishes to verify that the Key Curator accurately registered their public key, they do not need to download the entire system directory.

Instead, the user requests a *Merkle Proof of Inclusion*. This proof consists only of the specific sibling hashes required to trace a path from the user's leaf node up to the root. Because a binary tree halves the required data at every level, the user can mathematically prove their inclusion in the system directory by computing just $O(\log N)$ hashes. For example, in a system with one million registered users, a proof of inclusion requires roughly 20 hash values. This allows Reg-ABE to scale to enterprise levels entirely independently of the curator's trust.

Chapter 4

Pairing-Based Registered ABE

This chapter introduces the structural framework and system architecture of pairing-based Registered Attribute-Based Encryption (Reg-ABE), heavily referencing the design introduced by Hohenberger, Lu, Waters, and Wu [4]. By avoiding boolean circuits entirely and relying on algebraic operations and vector commitments, this protocol delivers a scalable framework that neutralizes the key escrow threat without relying on an omnipotent central authority.

4.1 Core Architecture and the Key Curator Model

In standard ABE, a central authority holds a Master Secret Key (MSK). In Reg-ABE, this absolute authority is replaced by a completely transparent **Key Curator**. The scheme operates over a one-time, trustless generation of a Common Reference String (CRS). Once the CRS is generated by a setup algorithm, the setup party goes offline, and the system relies entirely on the auditable curator. The framework is defined by the following tuple of polynomial-time algorithms:

- **Setup**$(1^\lambda, 1^N) \rightarrow$ **crs:** A randomized algorithm that outputs the public common reference parameters for a maximum of N users.
- **KeyGen**$(\mathbf{crs}) \rightarrow (pk_i, sk_i)$**:** Run locally by user i. Outputs a public key pk_i and a secret key sk_i. The curator never sees sk_i.
- **Register**$(\mathbf{crs}, pk_i, \mathbf{Attributes}\ x) \rightarrow$ **aux:** Users send their public key and certified attributes to the Key Curator.
- **Update**$(\mathbf{crs}, \mathbf{aux}) \rightarrow (\mathbf{mpk}, \{hsk_i\})$**:** The Curator deterministically aggregates new users, updating the Master Public Key (mpk) and generating public Helper Secret Keys (hsk_i) for each user.

- **Encrypt**$(\mathbf{mpk}, \mathbf{Policy}\ y, m) \rightarrow \mathbf{ct}$**:** Uses the global mpk and an access policy y to encrypt message m.
- **Decrypt**$(sk_i, hsk_i, \mathbf{ct}) \rightarrow m$**:** A user combines their local sk_i, the curator's public hsk_i, and the ciphertext to recover the message (if their attributes satisfy y).

4.1.1 Proof of Correctness

In cryptographic protocols, correctness is the mathematical guarantee that a legitimate user, following the established procedures, will successfully decrypt a valid ciphertext. To understand how correctness is achieved in Reg-ABE, the process can be broken down into the handling of blinding factors and pairing cancellations:

- **The Aggregation Problem:** In Reg-ABE, the Key Curator aggregates the public keys of all registered users by multiplying them together to form a single Master Public Key (MPK). Consequently, a ciphertext encrypted under this MPK is mathematically entangled with the "noise" (cross-terms) generated by the public keys of every other user in the system.
- **The Helper Key:** To enable decryption, the curator utilizes public information to compute the exact structure of this noise for a specific user. This specific noise-cancellation data is packaged into a Helper Secret Key (HSK) and published.
- **The Mathematical Cancellation:** During the decryption phase, the user applies the bilinear pairing function $e(\cdot, \cdot)$ to their locally generated secret key and the curator's HSK. Because pairings inherently allow exponents to multiply, the pairing operation mathematically aligns the user's secret with the aggregated noise. When these paired results are divided or subtracted, the noise perfectly cancels out, yielding the identity element 1.
- **The Result:** Once the cross-term noise vanishes, the user is left with a clean Linear Secret Sharing Scheme (LSSS) matrix evaluation. If the user's attributes satisfy the defined access policy, they can reconstruct the final blinding scalar α and successfully unlock the plaintext payload.

4.1.2 Security Notion

The security models for Reg-ABE define the capabilities of potential adversaries and the guarantees provided by the system. The primary concepts underlying the security of these schemes include the attack models, the trust assumptions regarding the curator, and resistance to collusion.

IND-CPA vs. IND-CCA Security

- **IND-CPA (Indistinguishability under Chosen-Plaintext Attack):** In this standard model, an adversary provides two equal-length messages to an encryptor, who randomly selects and encrypts one. The scheme is IND-CPA secure if the adversary cannot determine which message was encrypted with a probability significantly better than a random guess (50%).
- **IND-CCA (Indistinguishability under Chosen-Ciphertext Attack):** This represents a strictly stronger security guarantee. The adversary plays the same indistinguishability game but is additionally granted access to a decryption oracle. The adversary can query the oracle to decrypt any ciphertext except the challenge ciphertext itself. A Reg-ABE scheme achieving IND-CCA security ensures that even if an attacker actively manipulates ciphertexts to observe the system's behavior, the underlying plaintext of the target ciphertext remains secure.

The Honest-but-Curious Curator

Unlike traditional ABE where the central Private Key Generator (PKG) acts as a fully trusted entity capable of decrypting any file, Reg-ABE assumes the Key Curator is "honest-but-curious":

- **Honest Execution:** The curator faithfully follows the protocol, correctly aggregating public keys and generating the accurate Helper Secret Keys required for the system to function.
- **Curious Observation:** The curator may attempt to analyze the public keys, registered attributes, and ciphertexts to extract private data.

Because users generate their own secret keys (sk_i) locally and never share them with the curator, it remains cryptographically impossible for the curious curator to decrypt user files, completely eliminating the key escrow vulnerability.

Collusion Resistance

Collusion resistance is the most critical security requirement of any ABE system. Suppose a ciphertext is encrypted under the policy *"Cardiologist AND Pediatrics"*. User A possesses the attribute *"Cardiologist"*, while User B possesses the attribute *"Pediatrics"*. Neither user can decrypt the file individually.

In a collusion attack, User A and User B attempt to combine their secret keys to satisfy the policy jointly. Reg-ABE defends against this by binding each user's attributes to a unique, randomly sampled scalar during their local key generation phase (e.g., r_A for User A and r_B for User B). The LSSS matrix reconstruction explicitly requires all

attribute shares to be blinded by the exact same underlying randomizer. Because User A's keys are mathematically tied to r_A and User B's keys to r_B, their mathematical shares are fundamentally incompatible. Attempting to pair them together will output invalid noise rather than the correct reconstruction constant, causing the decryption to fail securely.

4.2 Vector Commitments and Merkle Tree Registration

One of the major features of Reg-ABE is achieving $O(\sqrt{N})$ or logarithmic registration times. To manage thousands of dynamic users, the Key Curator does not simply append public keys to a massive list. Instead, public keys are compressed into a digest using **Vector Commitments** and **Merkle Trees**. When a user registers, their public key pk_i is treated as a component of a larger mathematical vector. Using a pairing-friendly group $\mathbb{G}$, the curator computes a commitment $\mathcal{C}$ to the vector of all registered keys $x = (x_1, \ldots, x_N)$:

$$\mathcal{C} = \prod_{i \in [N]} h_i^{x_i} \tag{4.1}$$

To make updates efficient, these commitments form the leaves of a Merkle Tree. As pairs of users register, their commitments are hashed together (e.g., $h_{12} = H(h_1||h_2)$) up to a single Root Hash. This Root acts as the system's MPK. Anyone can audit the curator by requesting a Merkle proof to verify that their specific public key is mathematically included in the MPK root without requiring trust.

4.3 Slotted Reg-ABE and Powers-of-Two Mechanism

Constructing Reg-ABE directly is complex due to the dynamic, sequential nature of users joining the system. Thus, constructions employ an intermediate primitive called **Slotted Reg-ABE**. In a slotted scheme, the system is initialized with a fixed capacity of L users (slots), and the curator aggregates all L keys simultaneously. To transition from a static Slotted Reg-ABE to a Full, dynamic Reg-ABE (where users join sequentially), the system utilizes a **"Powers-of-Two" merging mechanism**. Instead of maintaining one continuously growing slotted tree, the curator maintains a forest of independent slotted ABE trees of sizes $1, 2, 4, 8, \ldots, \log N$.

Example of Cascading Merges: Suppose the system currently has 15 users. The forest contains full trees of size 8, 4, 2, and 1. When User 16 registers, they fill a new slot of size 1. This triggers a cascading collapse: the two trees of size 1 merge into a tree of size 2; those merge with the existing size 2 tree into a size 4, and so on, until the system has a single, perfectly balanced tree of size 16. This structure ensures that updates to the MPK and HSKs remain logarithmically bounded, providing efficiency gains over

traditional centralized PKGs.

4.4 Predicate Encodings and LSSS Policies

To enforce fine-grained access control, the pairing-based Reg-ABE maps boolean access formulas into Linear Secret Sharing Schemes (LSSS) using **Predicate Encodings**. Let $x \in \mathcal{X}$ be a user's attribute set, and $y \in \mathcal{Y}$ be the ciphertext's access policy. The encoding generates specific matrices and vectors over a finite field $\mathbb{Z}_p$:

- **Ciphertext Encoding (C_x):** Represents the attributes mathematically.
- **Key Encoding (K_y):** Represents the LSSS access matrix of the policy.
- **Decryption Vector ($d_{x,y}$):** A vector computable only if the predicate is satisfied ($P(x, y) = 1$).

During decryption, if the user's attributes satisfy the policy, they can perform the **α-reconstruction**. By arranging the matrices as $M_{x,y} = \begin{pmatrix} a_y & 0 \\ K_y & C_x \end{pmatrix}$, the user computes $M_{x,y} d_{x,y}^\top = e_1^\top$. This algebraic cancellation isolates the shared secret α, allowing the user to unblind the payload and recover the message. If $P(x, y) = 0$, the structure statistically hides α, guaranteeing information-theoretic security against unauthorized access.

4.5 Algebraic Aggregation and Cross-Terms

In pairing-based Reg-ABE, the actual aggregation of user keys occurs algebraically within the bilinear groups $(\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T)$ where $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$. A user i samples a local secret $r_i \leftarrow \mathbb{Z}_p$ and publishes $T_i = g^{r_i}$. When registered, the curator aggregates the keys via multiplication: $\hat{T} = \prod T_j$. Because the resulting MPK incorporates exponents from all users, a single user requires the **Helper Secret Key (HSK)** to filter out the "cross-terms" belonging to everyone else. The curator computes these cross-terms (e.g., blinding factors $V_{j,i} = A_j^{r_i}$) using only public information and packages them into hsk_i. During decryption, user i leverages the bilinear map to pair their private key with the HSK:

$$e(C, h_{N+1-i}) = e(\Lambda_i, g) \cdot e(h_i^{r_i}, h_{N+1-i}) \tag{4.2}$$

This pairing check effectively isolates their specific slot from the aggregated pool. Once the slot is verified and the cross-terms are canceled out, the user proceeds to the LSSS policy check to decrypt the final message.

Chapter 5

Efficiency Optimization via Prime-Order Groups

While composite-order configurations provide a very intuitive mechanism for separating data payloads and blinding factors into isolated subgroups, they carry a severe computational penalty. To keep the system secure, the factorization of the group order N must remain computationally impossible. This strict requirement forces cryptographic parameters to balloon to sizes between 1024 and 2048 bits. In contrast, prime-order elliptic curves can achieve the exact same security thresholds using a fraction of the space typically between 160 and 256 bits. By transitioning cryptographic operations into prime-order groups, systems benefit from significantly reduced ciphertext sizes, minimal memory footprints, and highly efficient pairing evaluations.

5.1 Replacing Numbers with Vectors

The core mechanic of composite-order Reg-ABE is "hiding" secrets within specific subgroups. Because operations in subgroup p_1 do not interact with operations in subgroup p_2, cryptographic blinding factors can be securely canceled out during decryption without interfering with the message payload. To achieve this same structural orthogonality in prime-order groups (which have no non-trivial subgroups), we must undergo a dimensional shift: **we replace numbers with vectors.** As formalized by Freeman (2010) [2] and heavily utilized in modern pairing-based cryptography, the transition maps composite properties to prime vector spaces as follows:

- **Group Element** (g) $\longrightarrow$ **Vector of size** d ($\vec{v}$).
- **Group Multiplication** ($g \cdot h$) $\longrightarrow$ **Vector Addition** ($\vec{v} + \vec{c}$).
- **Hidden Subgroups** ($\mathbb{G}_{p1}, \mathbb{G}_{p2}$) $\longrightarrow$ **Hidden Subspaces of a Vector Space** ($\mathbb{V}_1, \mathbb{V}_2$).

- **Standard Pairing** ($e(g, h)$) $\longrightarrow$ **Dot Product Pairings** ($e(\vec{v}, \vec{c}) = \prod e(v_i, c_i)$).

By taking a prime-order group $\mathbb{G}$ and constructing the product vector space $\mathbb{V} = \mathbb{G}^n$, we can simulate subgroup orthogonality. Subgroups in the composite setting are replaced by orthogonal subspaces, allowing the cancellation properties required for Reg-ABE decryption to function effectively.

5.2 Dual Pairing Vector Spaces (DPVS) and Simulating Composite Groups

To successfully implement this dimensional shift in a working public-key environment, state-of-the-art Reg-ABE protocols rely on Dual Pairing Vector Spaces (DPVS). Expanding upon Freeman's mathematical conversions, Lewko (2011) [5] developed a comprehensive toolkit that allows cryptosystems dependent on composite "cancellation properties" to function effectively within simulated prime vector spaces. Let $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ be an asymmetric pairing over prime-order groups. The setup algorithm randomly samples two orthogonal bases for the vector spaces:

$$\mathbb{B} = (b_1, b_2, b_3, b_4) \in \mathbb{G}_1^4 \tag{5.1}$$

$$\mathbb{B}^* = (b_1^*, b_2^*, b_3^*, b_4^*) \in \mathbb{G}_2^4 \tag{5.2}$$

These bases are meticulously constructed such that their dot-product pairings exhibit orthogonality. For a non-zero scalar $\psi \in \mathbb{Z}_p$, the pairing of the basis vectors satisfies:

$$e(b_i, b_j^*) = \begin{cases} g_T^{\psi} & \text{if } i = j \\ 1 & \text{if } i \neq j \end{cases} \tag{5.3}$$

In this DPVS framework, the Master Public Key (MPK) and ciphertexts live in the span of $\mathbb{B}$ (in $\mathbb{G}_1$), while the user secret keys and the curator's Helper Secret Keys (HSK) live in the span of the dual basis $\mathbb{B}^*$ (in $\mathbb{G}_2$). The cross-terms generated during the curator's aggregation phase are placed into specifically designated "blinding subspaces" (e.g., spanned by b_2, b_3). During decryption, the orthogonality mathematically guarantees that $e(b_{\text{data}}, b_{\text{blind}}^*) = 1$, vanishing the noise while preserving the secret α.

5.3 From Subgroup Hiding to the Decision Linear (DLIN) Assumption

In traditional composite-order cryptography, system security is anchored by the *Subgroup Decision Assumption*—meaning an adversary cannot discern whether a given element resides in the full group $\mathbb{G}_N$ or a constrained subgroup $\mathbb{G}_{p1}$. When these operations are migrated into prime-order vector spaces, this security baseline is replaced by the Decision Linear (DLIN) Assumption, and is often generalized further into the k-Linear (k-Lin) framework. Under the k-Lin assumption, given vectors in an affine subspace of dimension k, it is computationally intractable for any polynomial-time adversary to determine if a challenge vector lies within that specific subspace or if it is distributed uniformly at random across the entire vector space. If an adversary cannot distinguish the vector's subspace, they cannot differentiate between a valid ciphertext and random group elements without satisfying the access policy, thus guaranteeing semantic security.

5.4 Kronecker Products and QA-NIZK

As detailed by Zhu et al. (2023) [9], executing the linear algebra required to map LSSS access matrices across these vector networks relies heavily on the use of the Kronecker Product ($\otimes$) during the encryption routines. Assuming A represents the access matrix and B represents the basis matrix, calculating their Kronecker product $A \otimes B$ allows the encrypting party to merge the policy logic directly onto the DPVS basis vectors through a single tensor calculation:

$$A \otimes B = \begin{pmatrix} a_{11}B & \dots & a_{1m}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \dots & a_{nm}B \end{pmatrix} \tag{5.4}$$

Additionally, because prime-order spaces operate under highly constrained algebraic rules, modern slotted Reg-ABE protocols can easily incorporate Quasi-Adaptive Non-Interactive Zero-Knowledge (QA-NIZK) proofs. By appending a QA-NIZK proof π to the user's registered public key, the Key Curator can mathematically verify that the user's vector components are structurally sound and well-formed before aggregating them into the MPK, preventing malicious key-injection attacks without adding significant computational overhead.

5.5 Practical,Non-Monotone and CCA-Secure Reg-ABE

By carefully optimizing the sizes of these parameters and refining how LSSS encodings are projected within prime-order boundaries, Rouselakis and Tomida [6] engineered an highly space-efficient Reg-ABE framework. In their realistic parameter setting, the sizes of the MPK and the ciphertext are reduced to approximately 7.5KB. This represents a reduction of more than 2000x for the MPK and 5x for the ciphertext compared to previous state-of-the-art constructions. Beyond just shrinking the cryptographic footprint, this updated architecture introduces two critical enhancements required for real-world enterprise deployment:

1. **Non-Monotone Access Policies:** The system natively interprets negative logic. This allows encryptors to directly insert exclusionary rules into the matrix mapping, such as restricting access to personnel who are Faculty $\wedge$ $\neg$Contractor.

2. **CCA-Security:** The protocol achieves full Chosen-Ciphertext Attack (CCA) resilience, protecting the encrypted payload against aggressive attackers who might attempt to manipulate decryption oracles. This is a significant operational advancement, as older baseline schemes were generally limited to Chosen-Plaintext Attack (CPA) security.

By utilizing orthogonal vector projections in $\mathbb{Z}_p$ to simulate composite subgroup cancellations, alongside aggressive parameter tuning and DLIN/QA-NIZK security proofs, these cutting-edge models push Reg-ABE into highly practical territory, proving its readiness for modern cloud networks.

Chapter 6

More Reg-ABE and Functional Encryption

Building upon the robust foundation of Registered Attribute-Based Encryption (Reg-ABE), the current frontier of decentralized, escrow-free cryptography is **Registered Functional Encryption (Reg-FE)**. In standard ABE, decryption is strictly binary: a user either possesses the attributes to satisfy the policy and recovers the entire plaintext message, or they recover absolutely nothing. Functional Encryption (FE) is vastly more expressive. It allows a user's secret key, associated with a specific function f, to evaluate over an encrypted plaintext payload x and output exactly the computed result $f(x)$, revealing no other information about the underlying data x itself. This phase of the analysis reviewed the latest iterations of Reg-FE, specifically examining the advanced frameworks engineered by Zhu et al. [8], which are capable of executing complex operational tasks far beyond rudimentary Boolean filtering.

6.1 Registered Inner-Product Functional Encryption (Reg-IPFE)

The most fundamental building block of functional computation is the inner product. In a Reg-IPFE scheme, a user registers a linear function represented by a vector $\vec{y}$. The data provider encrypts a highly sensitive data vector $\vec{x}$ (e.g., financial metrics or medical vitals) under the system's Master Public Key (MPK). When the registered user attempts to decrypt the ciphertext, the cryptographic pairing operations do not yield $\vec{x}$. Instead, they mathematically isolate and output only the inner product:

$$f_{\vec{y}}(\vec{x}) = \langle \vec{x}, \vec{y} \rangle = \sum_{i=1}^{n} x_i y_i \tag{6.1}$$

In their prime-order group construction, Zhu et al. successfully engineered adaptive

Indistinguishability-security (IND-security) for the Reg-IPFE environment, anchored by the k-Linear (k-Lin) assumption. By integrating a complex nested dual-system mechanism within the security proof, the architecture aggressively curbs entropy leakage over sequential key queries. This refined structure results in weakly compact ciphertexts of size $O(n \cdot \log L)$, where n acts as the data vector dimension and L represents the total slot capacity of the system. This breakthrough enables highly secure, fully decentralized statistical processing over encrypted data streams without ever needing to place trust in a central authority.

6.2 Registered Quadratic Functional Encryption (Reg-QFE)

To push policy capabilities even further, this report also analyzed Reg-QFE, a protocol that allows registered users to evaluate quadratic polynomials over blinded data arrays. When a ciphertext securely houses two encrypted variables $(\vec{x}_1, \vec{x}_2)$, the designated decryption sequence deterministically calculates the following quadratic output:

$$f(\vec{x}_1, \vec{x}_2) = (\vec{x}_1 \otimes \vec{x}_2) f^\top \tag{6.2}$$

where $\otimes$ denotes the Kronecker operation. This allows for significantly more complex data analysis, such as computing variances or spatial distance metrics on encrypted datasets. This advanced construction achieves Selective Simulation-based Security (SIM-security) relying on the Bilateral k-Lin assumption.

6.2.1 The Pre-Constrained Registration Technique

A notable technical innovation introduced to make Reg-QFE mathematically feasible is the concept of **Pre-Constrained Registered IPFE (PReg-IPFE)**. Trying to compute complex polynomial functions natively across standard elliptic curves invariably generates non-linear terms. These non-linear elements fundamentally break the clean algebraic symmetry that the Key Curator depends upon to deterministically aggregate public parameters. To resolve this, the PReg-IPFE technique separates the function definition from the user registration. The master matrix defining the function space is embedded directly into the Common Reference String (CRS) during the one-time trusted setup. Consequently, users simply register standard, unconstrained integer vectors with the Key Curator. The ciphertext mathematics dynamically enforce the complex quadratic constraints during the decryption pairing operations. This separation ensures that the curator's workload remains purely linear (simple multiplications of group elements), allowing Reg-QFE to function efficiently in a completely decentralized environment.

Chapter 7

Conclusion and Future Directions

7.1 Conclusion

This project investigated the key-escrow-free cryptography domain. The preliminary phase of this research (Project-I) identified the severe computational roadblocks of non-black-box Identity-Based Encryption from standard CDH assumptions. By analyzing the substantial circuit overheads, we proved that literal garbled-circuit implementations are currently impractical for real-world deployment. This semester, by transitioning to **Registered Attribute-Based Encryption (Reg-ABE)** via pairing-based cryptography, we demonstrated that decentralized, escrow-free access control can indeed be achieved efficiently. We explored the theoretical translation of these schemes from composite-order groups into highly efficient prime-order vector spaces using the k-Linear assumption and DPVS techniques. Furthermore, we investigated the expansion of these protocols into Registered Functional Encryption (Reg-FE) frameworks, which permit advanced inner-product and quadratic computations directly on encrypted payloads. By replacing a centralized server with an auditable, deterministic Key Curator, Reg-ABE and Reg-FE represent the most viable and highly deployable fixes to the key escrow dilemma, a structural flaw that has plagued cloud cryptography for nearly two decades.

7.2 Open Problems and Next Plans

As this research progresses, several highly critical challenges remain to be solved:

- **Post-Quantum Reg-ABE (Lattice-Based):** Current practical constructions rely entirely on bilinear pairings over elliptic curves, which are fundamentally vulnerable to Shor's algorithm on quantum computers. Constructing fully efficient Reg-ABE from lattice-based assumptions such as Learning With Errors (LWE) without falling back on impractical Garbled Circuits remains a major open objective. My immediate

theoretical goal is to study lattice-based vector commitments to adapt the registration paradigm for the post-quantum era.

- **Compactness in Reg-QFE:** While pairing-based Reg-QFE is a significant step forward, current schemes suffer from a large Common Reference String of size $O(n^2 \cdot L^2 \cdot \log L)$. Reducing this CRS to a linear dependency on n and L, while maintaining full Simulation-based security, is a major open problem requiring novel algebraic techniques in dual-system encryption.

- **Traitor Tracing from Reg-ABE:** In a decentralized system, if a malicious user leaks their locally generated secret key or the curator's Helper Secret Key to an unauthorized third party, it is difficult to identify the source of the leak. Integrating efficient *Traitor Tracing* mechanisms into the Registered ABE framework—allowing the system to mathematically trace leaked decryption modules back to the specific registered slot—is a crucial next step for enterprise accountability.

- **Unbounded Scaling without iO:** Current pairing-based implementations enforce a rigid maximum capacity L for user slots. At present, removing this limitation relies entirely on Indistinguishability Obfuscation (iO), which provides strong theoretical guarantees but limited practical applicability. Discovering a methodology to allow infinite, dynamic user scaling using only standard black-box mathematics is a top priority.

- **Software Implementation and Benchmarking:** Transitioning away from pure theoretical proofs, my next immediate step is writing the prime-order Reg-IPFE logic into functional software. I plan to use highly optimized C/C++ libraries (like RELIC or PBC) to record actual setup, registration, and decryption latencies. This effort will establish reliable benchmarks measured against traditional escrowed ABE architectures operating under realistic cloud workloads.

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